

its 'hardness' and explains its importance observed in the case studies below. Among the multitude of complex relationships between many variables, it also provides a clear and hard boundary. Finally, it thereby reinforces the links between asset, basis and credit default swaps described in the previous chapters via fair value models with model-independent arbitrage flows.

Dealing with the arbitrage inequality has a few practical implications:

- The inequality allows arbitrage trades only if the USD ASW exceeds the CDS (i.e. only long asset&basis swapped bonds and long CDS positions).
- It therefore avoids the pitfalls of relying too much on the arbitrage equality in the presence of estimation risk, for example, the dangers of reverse asset&basis swapping bonds and selling CDS protection when the USD ASW is slightly below the adjusted CDS.
- Being long an asset&basis swapped bond and long the CDS is also the more attractive side for a number of practical reasons. In particular, in the event that an issuer nears default, repo markets may well cease to function altogether, which could make it simply impracticable to maintain a short position in the reverse asset&basis swapped bond versus a short position in the CDS. Moreover, on this side of the trade we do not need to worry about the bond becoming special in repo, being short a delivery option, or the FX component.
- On the other hand, the replacement of quantitative adjustments by qualitative assumptions weakens the sharpness of the arbitrage relationship. It is therefore probable that quite a number of violations of the arbitrage equality do not show up as violations of the weaker inequality between the traded instruments. As a result, some theoretical arbitrage opportunities cannot be executed in practice.
- One can improve the inequality by including a quantification for the SOFR–repo basis swap and the haircut, i.e. by refraining from the simplifications on the left side.
- In the conceptual discussion above, we have excluded coupon effects and technical issues (such as counterparty risk or liquidity). Thus, while the arbitrage inequality suggests that the CDS basis should always be positive, it is possible that there are negative bases due to coupon or technical effects not covered here.
- Given the capital charge for holding bonds, CCBS and CDS, sufficient availability of capital for arbitrage trades is a condition for the arbitrage inequality to be imposed on the market. In the presence of balance sheet constraints, violations of the arbitrage inequality (negative CDS bases) can be quite high and persist for some time. Actually, the extent of these violations is a good proxy for the scarcity of arbitrage capital. This is one topic of Chapter 18.

In order to monitor the market for violations of the arbitrage inequality, we recommend comparing the two spreads graphically against USD SOFR (formerly LIBOR), from asset&basis swapped bonds and from CDS for every issuer. This could be done, for example, by adding the CDS (ask) quotes as a line to the USD ASW levels shown as a function of maturity, as in Figure 16.1. Figure 16.3 provides this picture for Japan.

Note how well the no-arbitrage inequality is observed by the market. The USD ASW of JGBs with short time to maturity follow the upper boundary given by the CDS ask level very closely, while JGBs with a longer time to maturity quote significantly below the CDS ask level. This means that there are currently no trading opportunities for asset&basis swapped JGBs versus the Japanese CDS.⁹

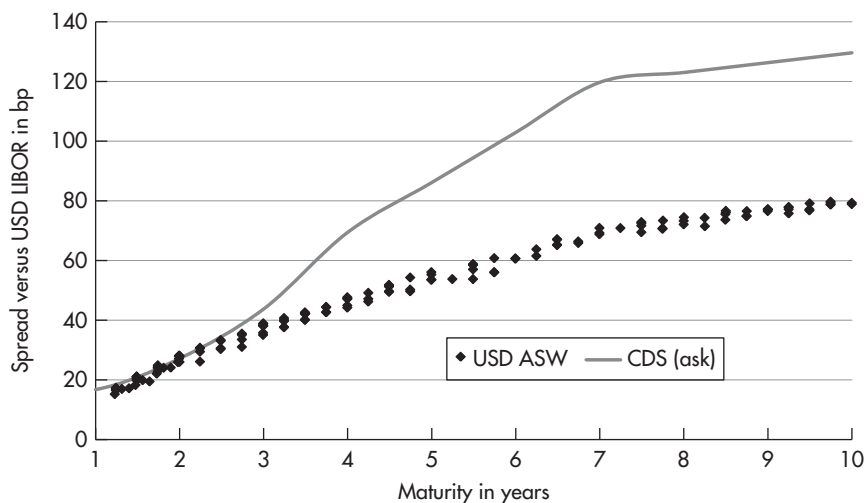


FIGURE 16.3 USD ASW of JGBs versus the Japanese CDS of the same maturity as the bonds.

Sources: data – Bloomberg; chart – Authors.

Data period: “Current” market data as of 23 Sep 2012.

⁹Buying 2Y asset&basis swapped JGBs and 2Y CDS on Japan at a basis of zero could be considered as buying the delivery option and the position on JPY weakening in case of Japan defaulting for free and therefore to be a good idea, though the challenges to estimate the value of this opportunity remain. Also, very short JGBs with time to maturity of less than one year (not shown in the chart) could actually have a USD ASW in excess of the CDS (if it traded for such short maturities.) (See Figure 16.1.) However, there could be special influences from redemptions distorting the ASW level.

In our experience, most of the violations of the no-arbitrage inequality occur in practice due to strong volatility in the CCBS, which can increase the USD ASW faster than the CDS (and faster than local ASW react). As a case study of the way trading opportunities from violations of the no-arbitrage inequality arise and how quickly they perform, we depict the history of the USD ASW of a 10Y Korean government bond versus the 10Y CDS ask level for Korea in Figure 16.4. Following the Lehman crisis, the KRW ASW of Korean government bonds widened, partly due to a general scarcity of risk capital forcing some investors to sell their bond holdings. At the same time, the KRW CCBS became highly negative as Korean banks needed USD funding. On the other hand, while the Korean CDS also widened, it followed a different timing (probably because the credit quality of the Korean government was not directly linked to the trouble in the US banking system). As a result, the USD ASW of Korean government bonds at times greatly exceeded their CDS levels, presenting those investors with risk capital available with arbitrage profits in excess of 100 bp. These precedents illustrate that constant monitoring of USD ASW through charts like Figure 16.3 can be very rewarding.

Figure 16.4 illustrates the frequency and magnitude of these violations of the arbitrage inequality, as well as the time over which these relative misvaluations have tended to correct. In quiet times, however, these opportunities

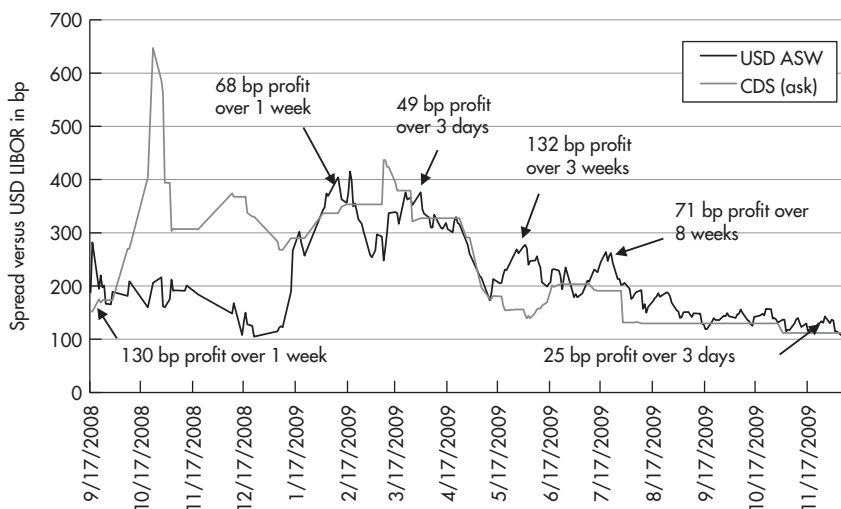


FIGURE 16.4 USD ASW of the Korean government bond 5.75% 09/18 versus the 10Y Korean CDS level.

Sources: data – Bloomberg; chart – Authors.

Data period: 17 Sep 2008 to 9 Dec 2009, trading daily data.

tend to be much rarer and to offer less profit. The shift to quiet times is visible at the right-hand side of the chart when the pricing of Korean bonds against USD LIBOR became closely linked to the Korean CDS level again. Following increasing regulatory and capital constraints, however, the threshold to engage in these arbitrage trades has increased for most market participants during the last few years as explained in Chapter 18.

Summarizing our experience, in comparison with the abundant relative value discovered by a principal component analysis on the CDS universe in Chapter 13, opportunities when the arbitrage inequality is violated arise infrequently. But when they do arise, they are usually good candidates for trades – as long as sufficient arbitrage capital is available. In contrast, bonds whose USD ASW spread is below the CDS are not necessarily candidates for similar trades. This side of the arbitrage would require both a way to quantify the necessary adjustments and an instrument to hedge against currency weakening post default. We have discussed above the difficulties involved. For the same reason, we do not generally recommend spread positions between two USD ASW versus CDS spreads between different maturities¹⁰ or different issuers.¹¹

As the arbitrage inequality introduces the addition information of a constraint, it can also support the global asset selection process pictured in Figure 16.1:

- For cases in which the USD ASW trades close to the CDS level, one can conclude that the credit risk (as priced in the CDS market) is fully reflected in the USD ASW level. Given the weakness of the inequality, the USD ASW could actually be too high relative to the CDS level, though it is hard to quantify by how much. Thus, incorporating Figure 16.3 back into the assessment of the USD ASW levels of JGBs from Figure 16.1, we can conclude that, unless the credit quality of Japan (as priced in the CDS market) deteriorates further, the USD ASW levels of 1Y and 2Y JGBs should not continue to increase.
- For cases in which the USD ASW trades considerably below the CDS level, on the other hand, the one-sidedness of the inequality prevents a firm conclusion. The USD ASW level of 79 bp for 10Y JGBs could be fair, too high, or too low compared to the 10Y CDS level of 130 bp.¹²

¹⁰For example, a “curve trade” long (asset&basis swapped) 2Y JGB/long 2Y Japan CDS against a short 7Y JGB (reverse asset&basis swapped)/short 7Y Japan CDS position (see Figure 16.3).

¹¹For example, a long (asset&basis swapped) JGB/long Japan CDS against a short US Treasury (reverse asset swapped)/short US CDS position (see Figure 16.1).

¹²While the relative value relationship provides little guidance, we could of course still consult fundamental credit analysis to address this issue.

- Taking the viewpoint of an investor allocating USD funds to JGBs, Figure 16.3 shows that the USD ASW of 10Y JGBs is still quite far below the CDS level. While we cannot say how far below the CDS level is “fair,” we can say that the USD ASW could increase by another 51 bp before the no-arbitrage inequality is violated. By contrast, the credit risk of Japan (as priced into the CDS) is at least fully reflected in the USD ASW of 2Y JGBs. Thus, in the event of stable CDS levels, the downside of basis-swapped 2Y JGBs is significantly smaller than for 10Y JGBs.

THE EQUILIBRIUM BETWEEN ASSET, BASIS, AND CREDIT DEFAULT SWAPS

At the beginning of this chapter, we described the equilibrium between all bond and swap markets globally given by the equation

$$USDASW = \frac{EURASW}{CF} - \frac{CCBS}{CF}$$

We then turned our attention to the arbitrage (in)equality between USD ASW and CDS. Finally, we shall now look at the relationship between both, in other words, at the impact of the arbitrage (in)equality on the equilibrium. Under the assumption *Adjusted CDS* $\approx$ *USD ASW*, which might be acceptable in case of SOFR-ASW, the equilibrium equation can be easily expanded to cover four variables (market observations):

$$CDS \geq \text{Adjusted CDS} \approx USDASW = \frac{EURASW}{CF} - \frac{CCBS}{CF}$$

This combined formula can serve as a guide through the complex relationships between all global bond and swap markets. We have just applied it to complement the asset allocation process based on Figure 16.1 with the CDS line from Figure 16.3 as additional constraint, i.e. information.

We shall now describe how the difference between low-risk and high-risk bonds results in two different equilibria:

- For low-risk bonds, both the level and the volatility of the CDS are low. Via the arbitrage (in)equality, this translates into rather stable USD ASW. The stability of USD ASW in the equilibrium means that moves and volatility in local ASW are mostly absorbed by moves and volatility in the CCBS and vice versa.

- For high-risk bonds, both the level and the volatility of the CDS are high. Via the arbitrage (in)equality, this translates into volatile USD ASW. If local ASW are stable, the equilibrium therefore links a volatile CDS and USD ASW with a volatile CCBS.

Based on this insight, we can explain the different relationships between bond and swap markets observed for Bunds and JGBs.

THE EQUILIBRIUM FOR BUNDS (LOW-RISK BONDS)

Because both Treasuries and Bunds are considered to have high credit quality, their CDS is rather low and stable, hence also their SOFR-ASW (see model of Chapter 12). Furthermore, the similar CDS levels for the US and Germany are in line with the fact that Bunds are considered reasonably close substitutes for US Treasuries with about the same safe-haven status. Thus, if the USD ASW of Bunds become too high or too low, either against USD SOFR on an absolute basis or relative to the USD ASW of US Treasuries, investors allocating their funds among low-risk bond markets through a process illustrated in Figure 16.1 usually increase or decrease their Bund holdings. This keeps the USD ASW of Bunds at a comparatively stable level and a comparatively stable spread versus US Treasuries.

The mathematical consequence of the relative stability of USD ASW and CDS of Bunds is that the equilibrium equation

$$USDASW = \frac{EURASW}{CF} - \frac{CCBS}{CF}$$

links a volatile CCBS with a volatile local ASW. In other words, most of the moves on the right-hand side of the equation cancel out, leaving the left-hand side rather constant.

Figure 16.5 shows that this describes the actual market quite well. When Bunds richened considerably in the EUR funding market, a constant CCBS would have led to Bunds quoting through US Treasuries on a USD ASW basis. This situation prompts reallocation flows from Bunds into US Treasuries (in USD) and thus a widening (more negative) of the EUR CCBS. The result is that some of the richness of Bunds in the local funding market is absorbed by the CCBS. In terms of volatility, the CCBS therefore has a moderating impact on the USD ASW, as it absorbs some of the local funding volatility through global investment flows.

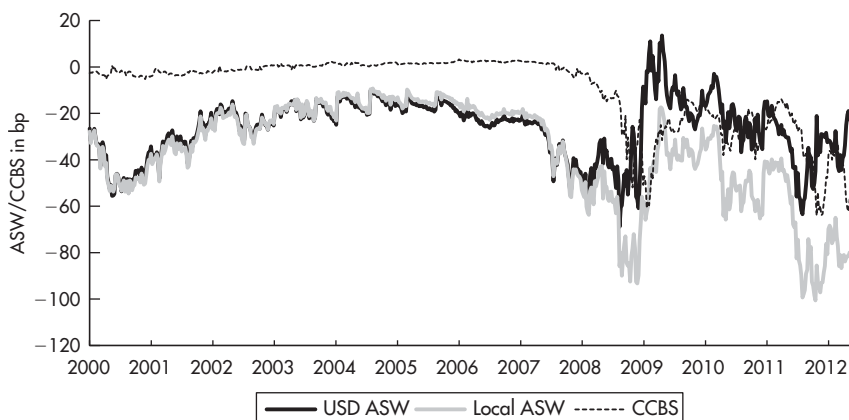


FIGURE 16.5 Local and USD ASW of 5Y Bunds versus the 5Y EUR CCBS.

Sources: data – Bloomberg; chart – Authors.

Data period: 8 Feb 2000 to 12 Jun 2012, weekly data.

THE EQUILIBRIUM IN CASE OF JGBS (RISKY BONDS)

Repeating the same analysis given in Figure 16.5 for Bunds with JGBs returns a completely different picture in Figure 16.6. Now the local ASW (rather than the USD ASW) is relatively stable and the USD ASW is a mirror image of the JPY CCBS. Thus, the same equilibrium equation between the three elements local ASW, USD ASW, and CCBS works in two different ways:

- In the case of Bunds, the USD ASW has little volatility, shifting most of the volatility from the CCBS to the local ASW (and vice versa).
- In the case of JGBs, the local ASW has little volatility, shifting most of the volatility from the CCBS to the USD ASW (and vice versa).

In the case of Bunds, the USD ASW is driven by the USD ASW of US Treasuries (Figure 16.1). By contrast, the USD ASW of JGBs is largely determined by the arbitrage (in)equality to the credit default swap (CDS) for Japan. As depicted in Figure 16.7, the correlation between the 5Y Japanese CDS and the USD ASW of 5Y JGBs between 2009 and June 2012 has been 0.78. Thus, the same widening of the CCBS in EUR and JPY (see Figure 15.3) was accompanied by two different effects:

- In the case of Bunds, a wider (more negative) CCBS correlates with lower local ASW (keeping the USD ASW rather stable) (Figure 16.5).

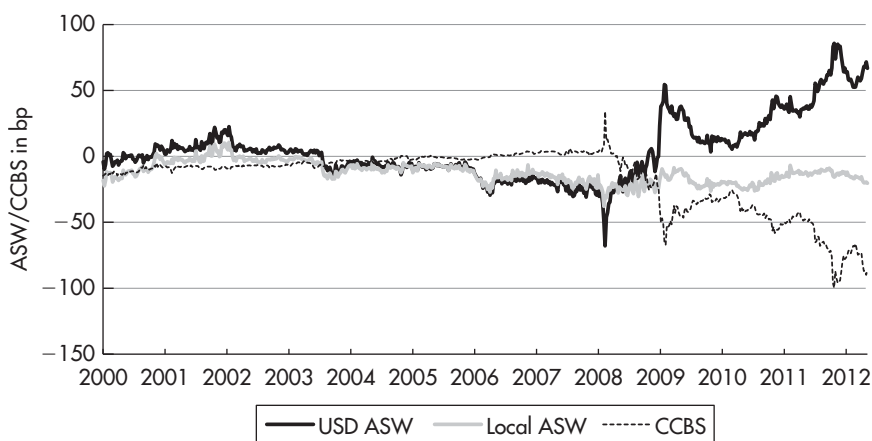


FIGURE 16.6 Local and USD ASW of 5Y JGBs versus the 5Y JPY CCBS.

Sources: data – Bloomberg; chart – Authors.

Data period: 8 Feb 2000 to 12 Jun 2012, weekly data.

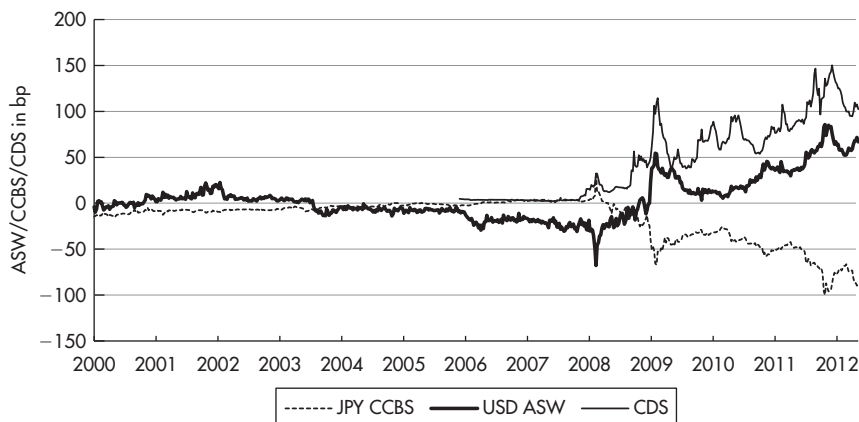


FIGURE 16.7 5Y JPY CCBS versus the USD ASW of 5Y JGBs and the 5Y Japanese CDS.

Sources: data – Bloomberg; chart – Authors.

Data period: 8 Feb 2000 to 12 Jun 2012, weekly data (CDS since 3 Jan 2006).

- In the case of JGBs, a wider (more negative) CCBS correlates with a higher USD ASW (keeping the local ASW rather stable) and thus a higher CDS premium (Figure 16.7).

Note that these relationships are symmetric and imply no statement about causality. It is quite possible that a higher Japanese CDS caused the wider JPY CCBS.

The combined formula has explained the behavior observed for Bunds by their similar credit standing to that of US Treasuries resulting in a rather tight and stable spread between the USD ASW of Bunds and US Treasuries and in a strong international participation in the Bund market. It can now explain the different behavior observed for JGBs by their different credit standing to that of US Treasuries. First, this leads to the USD ASW reflecting mainly the credit risk as expressed in a high and volatile Japanese CDS. Second, this limits the international participation in the JGB market. And with the JGB market remaining firmly in the hands of domestic investors, for whom the USD ASW may well be of little importance, there is no significant impact of the external factors USD ASW and CCBS on the local ASW and domestic funding conditions of JGBs. Hence, the same reason which makes USD ASW volatile tends to make local funding costs rather stable. And with local funding rates tending to be stable, the volatility of the CDS-driven USD ASW is mainly absorbed by CCBS volatility. This also provides a causal explanation for the relationship between the JPY CCBS and the USD ASW depicted in the regression chart in Figure 15.1.

Global Bond RV Via Fitted Curves and Via SOFR Asset Swap Spreads

INTRODUCTION

Chapter 16 has established the spread versus USD SOFR of asset&basis swapped bonds as universal yardstick, which expresses every bond globally in a single number, i.e. its SOFR-ASW. That chapter then went on to compare how one given bond is priced versus SOFR with the CDS of its issuer. This chapter will compare how different bonds are priced versus SOFR.

In fact, the SOFR-ASW seems to be an obvious choice to compare the richness/cheapness of all bonds worldwide: It is both a *universal* measure applicable to all bonds via asset&basis swapping and the *relevant* measure for global asset allocation decisions illustrated in Figure 16.1.

By contrast, LIBOR-ASW suffer from a number of deficiencies: due to the discrepancy between the LIBOR-swap curve and the bond yield curve, they are not suitable for comparing the relative richness/cheapness of different bonds.

The transition from LIBOR to SOFR, which has generally led to the convergence of future, swap and bond markets shown in Figure 12.3, resulted in a closer proximity between the SOFR-swap curve and the bond yield curve, hence in a better suitability of SOFR-ASW as a global RV indicator. At the core of this welcome development is the fact that the switch from LIBOR (and OIS) to SOFR eliminated the unsecured-secured basis from swap spreads. This offers the benefit of SOFR-swap spread models not depending on the unobservable cost of capital (Chapter 11) and hence an easy way to take the first hurdle toward the arbitrage equality by adjusting the swap spreads to reflect credit risk only (Chapter 16). And this chapter will exploit that the elimination of the unsecured-secured basis from swap spreads led to a convergence of swap and bond yield curves and therefore to an alleviation of the deficiencies of swap spreads as RV indicators.

However, in case swap and bond yield curves are not identical, there remain some problems in applying swap spreads as rich/cheap measures.